First Progress Review

Chapter 5: Work Plan and Timeline

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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5. Work Plan and Timeline

This chapter sets out a realistic plan for the 36-month, full-time programme that started on 1 October 2026 and ends on 30 September 2029. It explains the assumptions behind each duration, shows where the work stands at the time of writing (10 November 2026, month 2), and gives the Gantt chart, the dependencies between activities, a month-by-month plan to the First Progress Review, a quarterly plan for Years 2 and 3, milestones, the publication plan and the risk register. The plan revises the chart in the accepted proposal, which ran the registry, MRI, escalation and pathology work packages in parallel from month 5 to month 24 and left only the last eight months for federation, surgical planning, clinical validation and the thesis. For a single doctoral candidate that schedule was not achievable, so the work has been staggered: no more than 2 core research activities run at the same time, every activity that needs local data starts only after data access is expected, and two months of contingency are kept at the end.

5.1 Planning Principles

1. **Approvals first, and on the critical path.** Nothing involving patient data can start before ethics approval and a data-access agreement. These are scheduled realistically (Section 5.2) and all early research uses public data.

2. **Public data first, local data second.** Every model is first built and tested on public benchmarks (BraTS, BraTS-Africa, TCGA), so that the methods are ready when local data arrive and a delay in local data delays only the local experiments.
3. **At most two core research streams at a time.** One candidate cannot develop several models well in parallel. The work packages are staggered so that one stream is being developed while another is being written up.
4. **The core contribution is protected.** The escalation policy and orchestration (WP3, objectives O1 and O2) are completed by month 25. Federated deployment (WP5b) and the surgical-planning demonstrator (WP6) come later and can be narrowed without endangering the thesis.
5. **Each work package ends in a paper.** Papers are written as the work finishes, so most thesis chapters exist in draft before the final writing period.
6. **Contingency is explicit.** Months 35-36 are reserved for corrections and unexpected delays rather than planned work.

5.2 Assumptions Behind the Durations

Table 5.1: Duration assumptions and their basis.

Activity	Planned duration	Basis
KDU ethics review	3-4 months (M2-M5)	Submission in November 2026 with the committee's monthly meetings; allows one round of corrections
USJ ethics review	3-4 months (M5-M8)	Submitted after KDU approval, as advised by the supervisors on 5 November 2026
Hospital approval and data-access agreement	6-7 months (M6-M12)	Institutional approval, data-sharing agreement and IT access at NHSL; the slowest step in comparable studies, so it is started before USJ approval is complete
Registry build to pilot	8 months (M5-M15)	Requirements with clinicians, a minimum version (database, imaging server, de-identification, audit log) and a pilot; built on open-source components, not from scratch
Retrospective data collection and annotation	14 months (M14-M28)	Several hundred cases; MRI retrieval and de-identification in batches; clinician time is limited, so model-assisted annotation with review of a subset is used
Each model work package	6-8 months	Development, experiments, ablations and a paper draft, based on comparable doctoral and published studies
Clinical reader study	4 months data collection (M29-M32)	About 6-8 readers, short sessions around clinical duties, two reading conditions with a washout period
Thesis writing	6 months (M30-M35)	Chapters adapted from papers P0-P5 already drafted

5.3 Progress to Date

Table 5.2: Status of activities at 10 November 2026 (month 2).

Activity	Status	Evidence
PhD registration and proposal	Done	Proposal handed to FGS on 1 October 2026; accepted by the Board and handed to the evaluators on 4 November 2026
Literature collection	Done (first round)	Research Paper Bank of 559 open-access papers (O1 143, O2 109, O3 146, O4 161); 13 base papers; detailed methodology

Activity	Status	Evidence
		extracted for 246
Review tables	Done	One review sheet per objective, as instructed on 6 October 2026
Chapters 1, 2, 3 and 5 of this report	Draft	Version 1 written in November 2026 for supervisor review
KDU ethics application	Ready to submit	Signed by all investigators on 9 November 2026; REC fee paid on 10 November 2026 (receipt 13275)
USJ ethics application	Not started	After KDU approval
Progress website	Done	Updates, log book, timeline, documents and the Paper Bank for the supervisors
Preliminary results (Chapter 4)	Not started	Public-data MRI baselines planned for December 2026 - April 2027

5.4 Gantt Chart

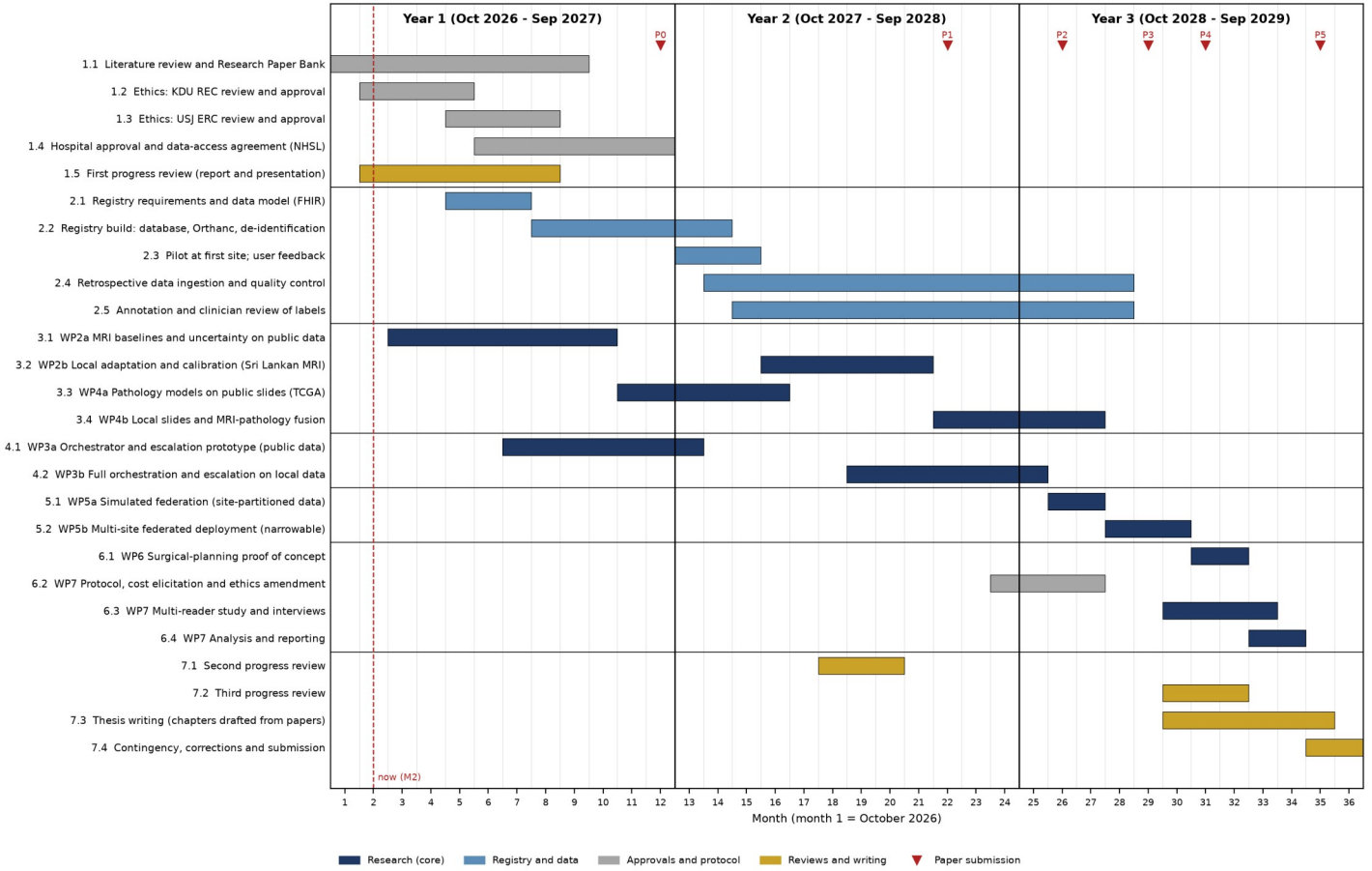


Figure 5.1: Gantt chart of the 36-month plan by month (month 1 = October 2026). Dark bars are core research, blue bars registry and data work, grey bars approvals and protocol work, and gold bars progress reviews and writing. Red triangles mark planned paper submissions; the dashed line marks the current month.

5.5 Activities, Dates and Dependencies

Table 5.3 lists every activity with its months, calendar dates and the activities it depends on. The critical path runs through ethics approval (1.2, 1.3), the data-access agreement (1.4), data ingestion and annotation (2.4, 2.5), local model adaptation (3.2), the full escalation study (4.2), the clinical reader study (6.3) and thesis writing (7.3). A delay on this path delays submission unless it is absorbed by the contingency months or by narrowing third-tier work.

Table 5.3: Activities, months, dates and dependencies.

No.	Activity	Months	Dates	Depends on
1.1	Literature review and Research Paper Bank	1-9	Oct 2026 - Jun 2027	-
1.2	Ethics: KDU REC review and approval	2-5	Nov 2026 - Feb 2027	-
1.3	Ethics: USJ ERC review and approval	5-8	Feb 2027 - May 2027	KDU approval (1.2)
1.4	Hospital approval and data-access agreement (NHSL)	6-12	Mar 2027 - Sep 2027	USJ approval (1.3)
1.5	First progress review (report and presentation)	2-8	Nov 2026 - May 2027	-
2.1	Registry requirements and data model (FHIR)	5-7	Feb 2027 - Apr 2027	-
2.2	Registry build: database, Orthanc, de-identification	8-14	May 2027 - Nov 2027	Data model (2.1)
2.3	Pilot at first site; user feedback	13-15	Oct 2027 - Dec 2027	Registry build (2.2)
2.4	Retrospective data ingestion and quality control	14-28	Nov 2027 - Jan 2029	Data access (1.4), pilot (2.3)
2.5	Annotation and clinician review of labels	15-28	Dec 2027 - Jan 2029	Ingestion (2.4)
3.1	WP2a MRI baselines and uncertainty on public data	3-10	Dec 2026 - Jul 2027	-
3.2	WP2b Local adaptation and calibration (Sri Lankan MRI)	16-21	Jan 2028 - Jun 2028	Local data (2.4, 2.5), baselines (3.1)
3.3	WP4a Pathology models on public slides (TCGA)	11-16	Aug 2027 - Jan 2028	-
3.4	WP4b Local slides and MRI-pathology fusion	22-27	Jul 2028 - Dec 2028	Pathology baseline (3.3), local slides (2.4)
4.1	WP3a Orchestrator and escalation prototype (public data)	7-13	Apr 2027 - Oct 2027	MRI baselines (3.1)
4.2	WP3b Full orchestration and escalation on local data	19-25	Apr 2028 - Oct 2028	Prototype (4.1), local models (3.2)
5.1	WP5a Simulated federation (site-partitioned data)	26-27	Nov 2028 - Dec 2028	Local models (3.2, 3.4)
5.2	WP5b Multi-site federated deployment (narrowable)	28-30	Jan 2029 - Mar 2029	Simulation (5.1), second site agreement
6.1	WP6 Surgical-planning proof of concept	31-32	Apr 2029 - May 2029	Local MRI models (3.2)
6.2	WP7 Protocol, cost elicitation and ethics amendment	24-27	Sep 2028 - Dec 2028	Escalation design (4.2)
6.3	WP7 Multi-reader study and interviews	30-33	Mar 2029 - Jun 2029	Full system (4.2, 3.4), amendment (6.2)
6.4	WP7 Analysis and reporting	33-34	Jun 2029 - Jul 2029	Reader study (6.3)
7.1	Second progress review	18-20	Mar 2028 - May 2028	-
7.2	Third progress review	30-32	Mar 2029 - May 2029	-
7.3	Thesis writing (chapters drafted from papers)	30-35	Mar 2029 - Aug 2029	Papers P1-P4
7.4	Contingency, corrections and submission	35-36	Aug 2029 - Sep 2029	Thesis draft (7.3)

The number of core research activities running at the same time never exceeds 2. In months when two run together, one is in development and the other is in its experimental or writing phase; for example, in months 22-25 the full escalation study (4.2) is being

completed while the pathology fusion work (3.4) starts on data that are already de-identified.

5.6 Detailed Plan to the First Progress Review (Months 2-8)

Table 5.4: Monthly plan, November 2026 to May 2027.

Month	Planned work	Output
Nov 2026 (M2)	Submit the KDU ethics application; supervisor review of Chapters 1, 2, 3 and 5; complete methodology extraction for the remaining Paper Bank papers	KDU submission; chapter comments
Dec 2026 (M3)	Revise chapters; set up the environment (PyTorch, MONAI, nnU-Net) on the available GPU; download and pre-process BraTS 2021 and BraTS-Africa	Chapters version 2; reproducible pre-processing
Jan 2027 (M4)	Train nnU-Net baseline on BraTS; add Monte Carlo dropout and a 5-member ensemble; measure Dice, HD95 and calibration; answer any KDU queries	First segmentation and calibration results
Feb 2027 (M5)	Test on BraTS-Africa as an external set; temperature scaling and reliability diagrams; registry requirements meetings with the clinical supervisor; prepare USJ ethics application after KDU approval	External-shift result; registry requirements
Mar 2027 (M6)	Uncertainty-error analysis and failure categories; registry data model (FHIR resources); start the NHSL data-access request	Failure taxonomy draft; data model
Apr 2027 (M7)	Write Chapter 4 (preliminary work and results); first simple escalation experiment (fixed vs calibrated threshold) on BraTS; assemble the full report	Chapter 4; full report draft
May 2027 (M8)	Supervisor sign-off; submit the First Progress Review (10 May 2027); presentation	Report and presentation

5.7 Plan for Years 2 and 3 by Quarter

Table 5.5: Quarterly plan after the First Progress Review.

Quarter	Main work	Expected outputs
Q3 Y1 (Apr-Jun 2027)	MRI baselines finished; escalation prototype on public data; registry build starts; USJ approval	Progress review 1; prototype
Q4 Y1 (Jul-Sep 2027)	Escalation prototype and cost-elicitation pilot; pathology baselines on TCGA; registry build; data-access agreement	P0 review submitted (M12)
Q1 Y2 (Oct-Dec 2027)	Registry pilot at the first site; ingestion and annotation start; pathology baselines finished	Registry live (M15)
Q2 Y2 (Jan-Mar 2028)	Local MRI adaptation and calibration; annotation continues; second progress review prepared	Local MRI results
Q3 Y2 (Apr-Jun 2028)	Full orchestration and escalation on local data; local MRI paper	Progress review 2 (M20); P1 (M22)
Q4 Y2 (Jul-Sep 2028)	Escalation study continued; MRI-pathology fusion on local slides; WP7 protocol and amendment	Fusion results
Q1 Y3 (Oct-Dec 2028)	Escalation study completed; fusion completed; simulated federation	P2 flagship (M26)
Q2 Y3 (Jan-Mar 2029)	Federated deployment; reader study starts (M30); thesis writing starts	P3 (M29)
Q3 Y3 (Apr-Jun 2029)	Reader study; surgical-planning demonstrator; federation paper; progress review 3; thesis writing	P4 (M31); progress review 3
Q4 Y3 (Jul-Sep 2029)	Reader-study analysis and clinical evaluation paper; thesis completion; contingency	P5 (M35); thesis submitted (M36)

5.8 Milestones

Table 5.6: Milestones.

No.	Milestone	Month	Date
M1	Proposal accepted	2	Nov 2026
M2	KDU ethics approval	5	Feb 2027
M3	First progress review	8	May 2027
M4	USJ ethics approval	8	May 2027
M5	Data access agreed	12	Sep 2027
M6	Registry live at first site	15	Dec 2027
M7	Second progress review	20	May 2028
M8	Escalation policy evaluated	25	Oct 2028
M9	Third progress review	32	May 2029
M10	Clinical validation complete	34	Jul 2029
M11	Thesis submitted	36	Sep 2029

5.9 Publication Plan

Table 5.7: Planned papers and submission months. Target venues are indicative and will be agreed with the supervisors.

No.	Paper	Submission	Indicative venue
P0	Review paper from the literature review	M12 (Sep 2027)	Artificial Intelligence in Medicine / npj Digital Medicine
P1	Calibrated, data-efficient MRI models (WP2)	M22 (Jul 2028)	MICCAI / Medical Image Analysis
P2	Calibrated selective autonomy across agents (WP3, flagship)	M26 (Nov 2028)	npj Digital Medicine / Nature Communications
P3	Pathology-MRI fusion with missing sources (WP4)	M29 (Feb 2029)	IEEE Transactions on Medical Imaging
P4	Federated learning and calibration (WP5)	M31 (Apr 2029)	Radiology: Artificial Intelligence
P5	Clinical evaluation of the HITL framework (WP7)	M35 (Aug 2029)	The Lancet Digital Health / JAMIA

A surgical-planning paper (WP6) is optional. At least four peer-reviewed outputs are expected, as committed in the proposal. If P4 or P5 is not accepted before submission, the thesis chapters stand on the submitted manuscripts.

5.10 Supervision and Reporting

- Principal supervisor (Dr. WMKS Ilmini): meetings every two weeks on research progress and writing.
- Co-supervisor (Dr. WGCW Kumara): monthly meetings on technical decisions, computing and experiments.
- Clinical co-supervisor (Dr. AM Deepal K Attanayake): monthly meetings on data access, error costs, annotation and the clinical study.
- Every meeting is recorded in the log book on the progress website, with decisions and actions; the timeline on the website follows this chapter and is updated monthly.

- Progress reviews to the Faculty of Graduate Studies at about months 8, 20 and 32, or as the Faculty schedules them.

5.11 Risks, Early Warnings and Fall-Back Plans

Table 5.8: Risk register.

Risk	Early warning	Response
Ethics approval takes longer than planned	No KDU decision by M5	Keep working on public data (no change to Year 1 research); move 1.4 and 2.4 back by the delay; use contingency months
Data-access agreement delayed	No agreement by M12	Start ingestion with a smaller retrospective set at the clinical supervisor's unit; run local experiments on the cases available; extend public-data experiments
Fewer usable cases than expected	Fewer than 150 usable MRI cases by M21	Rely more on transfer and self-supervised learning; report confidence intervals honestly; use BraTS-Africa as a second external set
Slides cannot be digitised at scale	Fewer than 80 scanned slides by M24	Use slides already scanned; keep WP4b to a feasibility study; fusion still evaluated on TCGA
Clinician time is scarce	Annotation below 10 cases a week	Model-assisted annotation with clinician correction; double annotation for a subset only; shorter reader sessions
Second hospital does not join	No second agreement by M27	Keep WP5 to simulated federation (5.1); this does not affect O1, O2 or the core of O3
Computing limits	Training runs exceed one week	Smaller configurations, frozen pathology encoders, mixed precision, university or cloud GPU time
Overall delay	More than two months behind at a progress review	Drop WP6 to a short demonstration and WP5b to simulation; keep WP3 and WP7, which carry the thesis

Minimum thesis. If several risks occur together, the thesis still stands on O1 and O2 (the orchestrator and the calibrated escalation policy) evaluated on public data and the available local cases, O3 for MRI and pathology on local data without multi-site federation, and O4 as a smaller reader study. This is the scope the plan protects first.